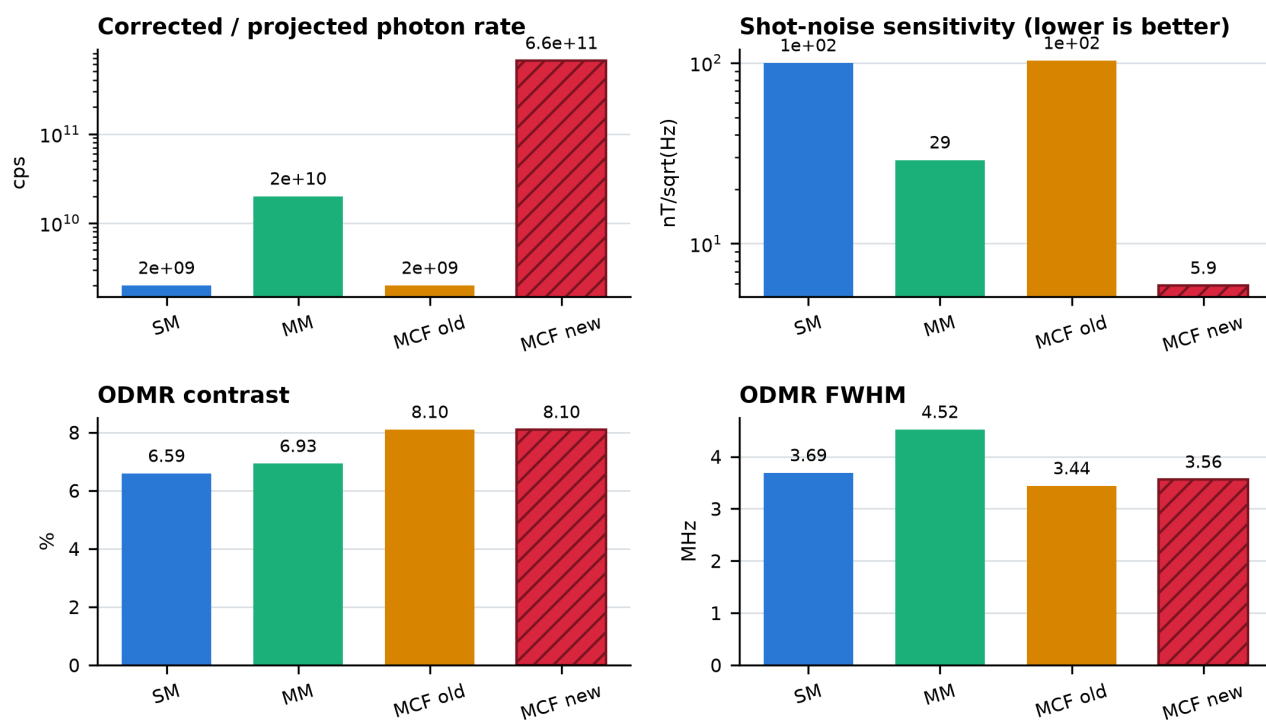


SM vs MM vs MCF optical probe comparison

Measured SM, MM and original MCF performance compared with the new monolithic seven-lens MCF model

Evidence boundary: SM, MM and MCF old are measured experimental values. MCF new is a shot-noise-limited optical-model prediction and has not yet been experimentally validated. Raw fiber photons and the empirical comparison normalization are reported separately.



Solid bars: measured experiment. Hatched bar: empirically normalized model target.

Primary comparison

Probe	Evidence	Photon rate*	Contrast	FWHM	Sensitivity
SM	Measured	2e+09 cps	6.59%	3.69 MHz	100.00 nT/sqrt(Hz)
MM	Measured	2e+10 cps	6.93%	4.52 MHz	29.00 nT/sqrt(Hz)
MCF old	Measured	2e+09 cps	8.10%	3.44 MHz	103.00 nT/sqrt(Hz)
MCF new	Normalized model	6.63e+11 cps	8.10%	3.56 MHz	5.86 nT/sqrt(Hz)

* Measured rates are corrected for ND attenuation. The new-MCF entry is a comparison-normalized count-rate proxy, not a physical detector conversion. Its separate raw output is reported below.

Main result

Under the explicitly labeled empirical normalization, the new design gives a 331.6x count-rate proxy and 17.6x lower shot-noise sensitivity than the measured original MCF, while retaining the measured 8.10% MCF contrast. The physical optical-model output is 6.05×10^{10} fiber photons/s, corresponding to 19.41 nT/sqrt(Hz) if one fiber photon is treated as one detected count. The measured MM remains the best demonstrated sensitivity at 29 nT/sqrt(Hz).

New MCF design and optical model

Printed structure	One monolithic IP-S body ($n = 1.52$): central lens plus six side lenses on a common support
Central channel	532 nm excitation; quadratic polynomial surface; 20 μm aperture; 120 μm protrusion
Side channels	650-850 nm collection; six biconic polynomial surfaces; 24 μm aperture; 35 μm core pitch; 10 μm MFD
Best working gap	5.0 μm central-lens-to-diamond clearance for the fixed optimized tip
NV ensemble	3 ppm NV centers throughout the 80-90 μm layer in a 100 μm diamond
Optical physics	3-D rays, local surface normals, vector Snell law, each Fresnel interface applied once, wavelength-dependent diamond index, saturation and excitation-collection overlap
Final quadrature	33 NV-depth planes and 33 trapezoid-weighted wavelengths

Complete monolithic MCF lens and 3-D Snell-law rays

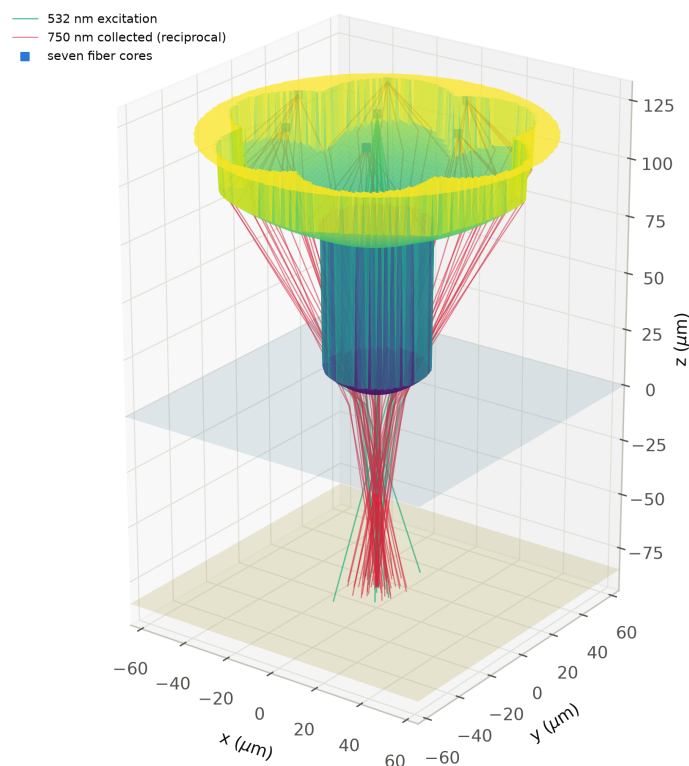


Figure 2. Complete seven-core printed-lens union. Green paths are 532 nm excitation rays. Red paths are reciprocal 750 nm collection rays. Refraction occurs only at the polymer-air and air-diamond boundaries.

Sensitivity model

Both outputs use $\eta_{\text{new}} = 103 \times (\text{FWHM}_{\text{new}} / 3.44) \times (0.081 / C_{\text{new}}) \times \sqrt{2e9 / R_{\text{new}}}$, with η in $\text{nT}/\sqrt{\text{Hz}}$. For the raw optical result, R is $6.05e+10$ fiber photons/s. For comparison with the measurements, R is the explicitly normalized proxy $6.63e+11$ cps. The normalization factor is 10.97; because it exceeds one, it is not interpreted as detector efficiency.

Mounting tolerance and interpretation

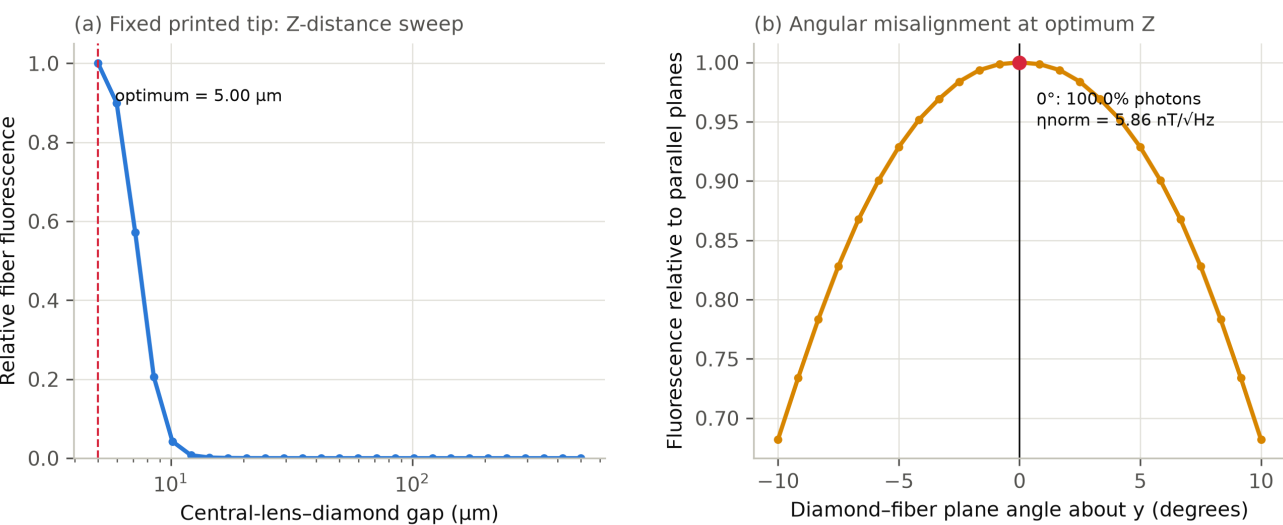


Figure 3. Fixed-lens Z-distance response and diamond-plane angular response. At nonzero tilt, all six side lenses are traced independently; the diamond and the 80-90 μm NV layer rotate together about the fiber y-axis.

Z sensitivity	The sampled fixed-tip optimum is 5.00 μm. At the sampled point nearest 10 μm (10.15 μm), modeled fiber fluorescence is 4.3% of the peak.
Angular sensitivity	At the sampled angles 5.00 and 10.00 degrees, predicted fiber-photon retention is 92.87% and 68.21%, respectively.
Measured ordering	SM and old MCF both give about 2e9 corrected cps. MM gives about 2e10 corrected cps and is therefore 10x higher in the supplied experiment.
Raw optical output	6.05e+10 fiber photons/s and 19.41 nT/sqrt(Hz) under the one-photon/one-count convention.
Normalized comparison	6.63e+11 cps proxy, 8.10% contrast, 3.564 MHz FWHM and 5.86 nT/sqrt(Hz).

Limits of the comparison

The new-design fiber-photon rate is an ideal smooth-surface geometrical-optics projection. The separate empirical normalization references the legacy old-MCF model checkpoint and must not be read as detector efficiency. The model does not include wave-optical diffraction at detailed freeform features, detector saturation or dead time, fabrication roughness and shrinkage, alignment drift, real wavelength-dependent core modes, fluorescence background, filter transmission or detector quantum efficiency. No background-derived contrast penalty is applied because no measured background-versus-footprint calibration was supplied.

Accordingly, compare the new MCF with the measured probes as a design target and expected trend, not as a demonstrated experimental result. The next decisive test is to fabricate the full STL, measure the unattenuated linear photon range, and repeat contrast/FWHM/sensitivity measurements at controlled gap and tilt.

Reproducibility

Inputs: figures/mcf_freeform_design.json and figures/mcf_gap_angle_sweep.json. Geometry: figures/mcf_freeform_full_seven_core.stl. Regenerate design figures with 'python redesign_fig.py --angle-deg ANGLE'. Generate this report with 'python make_comparison_pdf.py'.

Lens replication drawing

Use the full seven-core STL as the authoritative fabrication geometry. The drawing below is for alignment and metrology; do not reconstruct the surface from the drawing alone.

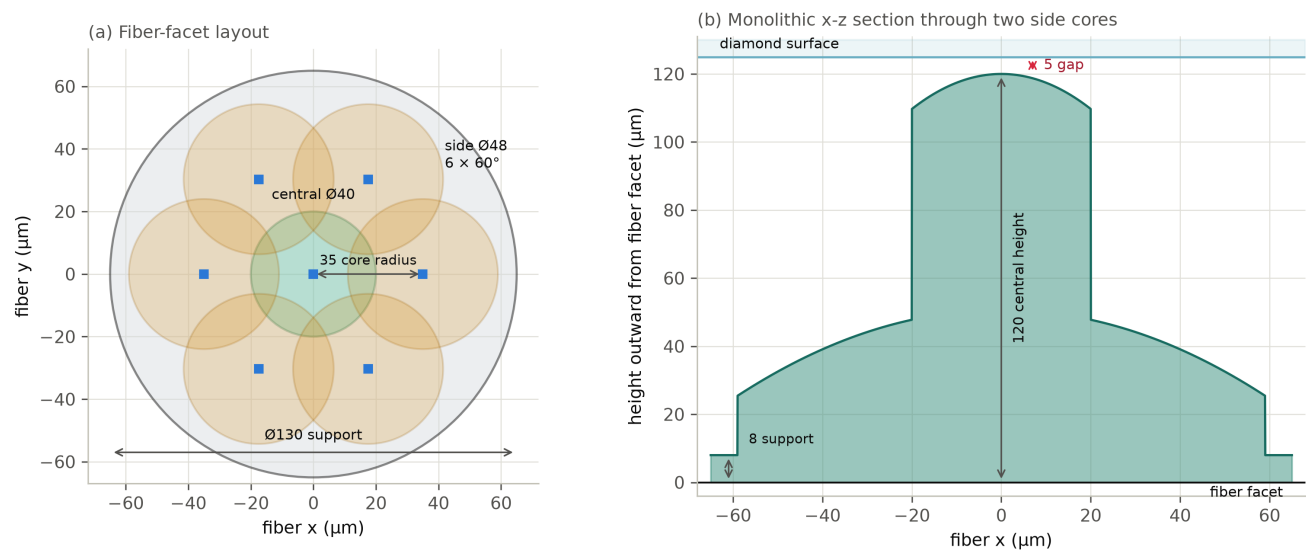


Figure 4. Fabrication coordinates use the fiber facet as $Z_f = 0$ and positive Z_f outward toward the diamond. Transparent aperture circles show where lens domains overlap; the printed result is one continuous height union.

Authoritative file	figures/mcf_freeform_full_seven_core.stl; binary STL; 31,104 triangles; one watertight body
STL units	The STL is unitless but every coordinate is in micrometers. Select um on import. If the CAD system assumes millimeters, apply scale 0.001.
Fiber coordinates	Central core at (0,0). Six side cores at radius 35 um and azimuths 0, 60, 120, 180, 240 and 300 degrees. All cores have 10 um MFD.
Support	Circular footprint diameter 130 um; 8 um support thickness at the fiber facet.
Central lens	Centered on the central core; aperture diameter 40 um; maximum outward height 120 um.
Side lenses	Six rotated copies; aperture diameter 48 um; aperture centers coincide with the six side cores.
Assembly spacing	Place the diamond surface 125 um from the fiber facet, giving a 5 um gap from the 120 um central tip. The NV layer is then 80-90 um inside the diamond.
Overlap rule	All lens and support domains are fused as one polymer body. Where surfaces overlap, keep the most outward surface; never add their heights and never preserve an internal optical boundary.

Exact CAD surface definition

This equation reproduces the same monolithic height field used by the STL generator. Distances are in micrometers. The STL remains preferable because it also fixes tessellation, footprint and watertight closure.

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For each lens, xi = u/a and nu = v/a, with u^2 + v^2 <= a^2.
S(u,v) = a*[p0*xi + 0.5*p1*xi^2 + 0.5*p2*nu^2
          + (p3/3)*xi^3 + p4*xi*nu^2 + (p5/4)*xi^4
          + 0.5*p6*xi^2*nu^2 + (p7/4)*nu^4] + shift
Hj(u,v) = base_z - apex_j - S(u,v)
H(x,y) = max(8, Hcentral, Hside_0, ... Hside_5)
Printed volume: x^2+y^2 <= 65^2 and 0 <= Zf <= H(x,y).
```

Coordinate transform	The simulation uses $z_model = 125 - Z_f$. The fiber-facet plane is $z_model = 125$; the diamond-facing direction is decreasing z_model .
Central local frame	$u=x, v=y; a=20; apex=5; base_z=125; shift=0$.
Central coefficients	$p0...p7 = [1.81356705e-17, 1.029120298, 1.029120298, 0, 0, 0, 0, 0]$
Side local frames	For $k=0...5$, $\phi=k*60$ deg, $center=35*(\cos(\phi), \sin(\phi))$; $u=(r-center).(\cos(\phi), \sin(\phi)); v=(r-center).(-\sin(\phi), \cos(\phi))$.
Side parameters	$a=24; apex=75; base_z=125; shift=8.17805531462$.
Side coefficients	$p0...p7 = [0.5073450157, 0.3544457901, 0.4666664528, 0, 0, -0.009099861824, -0.009099861824, -0.009099861824]$

Artifact identity

Full STL SHA-256: 781C6AA03572536BAF430F24A6E3A04BC23E6216E804B94A71E9555C399E82DB

Prototype file figures/mcf_freeform_central_one_side.stl contains only the central lens and one side domain. Use the full seven-core file for fabrication.

Important orientation check

Before slicing, display the fiber attachment plane and the lens tips in the machine preview. The flat disk at $z_model=125$ attaches to the fiber; the central tip is at $z_model=5$ and must project away from the fiber. The original job used InvertZAxis 1, but that command is machine-coordinate dependent and must not replace a visual orientation check.

Fabrication and verification procedure

The geometry is complete; the exposure recipe is not universally transferable. Follow this sequence and calibrate the printer/polymer combination with a coupon before writing on the MCF.

1. Verify the fiber

Image the facet and fit the seven core centers. Confirm one central core and six cores on a 35 μm radius. Use the measured central core as the $x=y=0$ registration point; use two or more side cores to determine rotation and scale.

2. Import and register the full STL

Import in micrometers, preserve aspect ratio, and align the STL central axis to the central core. Rotate about that axis until every side-lens center coincides with its side core. The support disk may be cropped only after confirming it remains fully attached to the actual fiber facet.

3. Slice as one monolithic solid

Union the geometry before slicing. Do not slice seven independent lenses and do not reuse the old AddZOffset 134.04 or 24.77 commands: the new STL already contains the complete relative heights. Start from the paper's 100 nm slice and 200 nm hatch settings, confirm them with an IP-S shrinkage coupon, then inspect the toolpath for uninterrupted support-to-lens attachment and no duplicate exposure in overlap regions.

4. Reported paper and DeScribe starting settings

Printer/objective	Nanoscribe Photonics Professional GT; 25X objective
Photoresist	IP-S photopolymer; paper reports n approximately 1.5
Slicing/hatching	100 nm slicing distance; 200 nm hatching distance
Exposure/delay	Laser power 33%; 3 ms delay after every written layer
Writing mode	GalvoScanMode; ContinuousMode
Speeds	Support ScanSpeed 50000; lens ScanSpeed 10000
Stage/control	PiezoSettlingTime 10; GalvoAcceleration 10; StageVelocity 200; PowerScaling 1.0
Original orientation	InvertZAxis 1

The printer, objective, slice, hatch, power and inter-layer delay are reported in the supplied paper. The remaining controls and scan speeds come from Cylinder_lenses_job.gwl. LaserPower 33% is still instrument-specific and must be verified for the actual objective, resin batch, interface and laser calibration.

5. Develop, inspect and mount

Use a validated IP-S development, rinse and drying procedure and record the exact solvents and times; the supplied paper states only that unpolymerized material was removed in solvent and the structure was dried. Inspect for delamination, voids and stair-stepping. Measure the 130 μm footprint, 8 μm support, 120 μm central height, 40 μm central aperture, 48 μm side apertures and 35 μm side-core radius. Compare a profilometer or confocal surface scan directly with the STL, especially across overlap seams.

6. Optical acceptance test

With the diamond parallel to the fiber, start at a 5 μm central-tip gap. Sweep Z before comparing count rates: the model falls to about 90% at 5.97 μm and about 4.3% near 10.15 μm . Verify central-core 532 nm delivery first, then inject or collect 650-850 nm light through each side core separately before summing all six.

Missing process data

The supplied paper does not identify the commercial MCF part number or cladding diameter, nor the development solvent/time, interface-finding method, shrinkage compensation, acceptable surface roughness or calibrated absolute laser dose for this new STL. Record those values from the actual fiber and a successful calibration coupon before another laboratory attempts process-level replication. The CAD geometry itself is fully specified here.